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Architecture deep dive: the model, the recipe, the evidence

This is the reference for teaching the project. It goes layer by layer through the model, derives the loss, explains the evaluation, lists every experiment with its numbers, and maps each idea to the paper it comes from.

- Start with how_it_works.md for the overview (task, data, big picture). This document assumes it and goes deeper.
- Code references are to commit `0e9ff1f` (2026-10-20) on branch `ws-m1-infra` (§9: `d3ed4ea`). Function and class names are stable; line numbers drift, so search for the name.
- Every number below comes from a file in the repo (`RESULTS.md`, `logs/`, `submissions/*/run_config*.json`) or from a script run on the model. Anything that couldn't be checked is marked **(unverified)**.

Contents

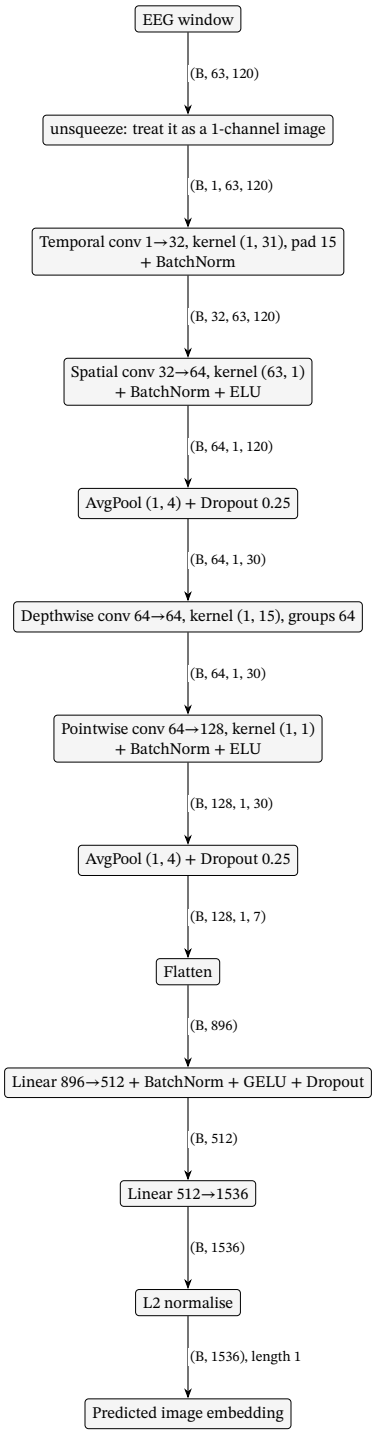
1. The encoder, layer by layer
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9. Next: new people and new headsets (in progress)

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1. The encoder, layer by layer

`EEGToImageEncoder` in `submission/submission.py`. One input window is a matrix of **63 electrodes × 120 time samples** (1 s at 120 Hz, from −0.2 s to +0.8 s around image onset). The output is one **unit-length 1536-d vector** in DINOv2-giant's embedding space.

1.1 The whole network at a glance



1.2 Layer table (computed, not estimated)

Shapes and parameter counts were produced by running the class on a `(2, 63, 120)` input on CPU (script in exercise 1, §8.3).

#	Module (exact PyTorch)	Shape in → out	Parameters
1	`Conv2d(1, 32, kernel_size=(1, 31), padding=(0, 15), bias=False)`	(B,1,63,120) → (B,32,63,120)	992
2	`BatchNorm2d(32)`	same	64
3	`Conv2d(32, 64, kernel_size=(63, 1), groups=1, bias=False)`	(B,32,63,120) → (B,64,1,120)	**129,024**
4	`BatchNorm2d(64)` · `ELU()`	same	128
5	`AvgPool2d((1, 4), stride=(1, 4))` · `Dropout(0.25)`	(B,64,1,120) → (B,64,1,30)	0
6	`Conv2d(64, 64, kernel_size=(1, 15), padding=(0, 7), groups=64, bias=False)`	(B,64,1,30) → same	960
7	`Conv2d(64, 128, kernel_size=(1, 1), bias=False)`	(B,64,1,30) → (B,128,1,30)	8,192
8	`BatchNorm2d(128)` · `ELU()`	same	256
9	`AvgPool2d((1, 4), stride=(1, 4))` · `Dropout(0.25)`	(B,128,1,30) → (B,128,1,7)	0
10	`flatten`	(B,128,1,7) → (B,896)	0
11	`Linear(896, 512)`	(B,896) → (B,512)	**459,264**
12	`BatchNorm1d(512)` · `GELU()` · `Dropout(0.25)`	same	1,024
13	`Linear(512, 1536)`	(B,512) → (B,1536)	**787,968**
14	`F.normalize(p=2, dim=-1)`	same	0
	Total		**1,387,872**

Where the parameters are: the convolutional "feature extractor" (layers 1–9) has 139,616 parameters (10%); the projection head (11–13) has 1,248,256 (90%). The network is mostly a mapping from 896 EEG features to DINOv2 space. The single biggest conv layer is the spatial one (129k), because every output filter sees all 32 temporal maps on all 63 electrodes.

Size variants from #27 (`temporal_filters`, `spatial_filters`, `sep_filters`, `hidden_dim`, `head_blocks`), also computed: wider convs 64/128/256 → 2.26M; head 1024 + 1 residual block → 5.00M; both → 6.33M.

1.3 What each layer does, and why

Input reshaping. EEG is not an image: one axis is space (electrodes), the other is time. Treating it as a 1-channel image `(1, 63, 120)` lets us use 2-D convolutions whose kernels are 1-D in one axis: `(1, k)` kernels act **along time only**, `(63, 1)` kernels **across electrodes only**. This "factorised" spatio-temporal design is the core idea of ShallowConvNet / DeepConvNet (Schirrmeister et al., 2017) and EEGNet (Lawhern et al., 2018).

1–2. Temporal convolution (32 filters × 31 taps) + BatchNorm.

- Each filter is a **learned FIR filter** applied identically to every electrode (the kernel is `(1, 31)`, so it slides along time and is shared across rows).
- 31 taps at 120 Hz span **258 ms**. A filter of that length can isolate frequency bands about $120/31 \approx 4$ Hz wide anywhere between 0 and the 60 Hz Nyquist limit: theta (4–8 Hz), alpha (8–13 Hz), beta, or act as a matched filter for the shape of an evoked response.

- `padding=15` keeps the 120 samples ("same" convolution). The first and last 15 samples are computed against zero padding, so the window edges are slightly less reliable.
- `bias=False` because the BatchNorm right after has its own shift.
- **BatchNorm2d** (Ioffe & Szegedy, 2015) standardises each filter's output over the batch, electrodes and time, which keeps training stable at a high learning rate.
- Lineage: EEGNet uses 8 filters of half the sampling rate (64 taps at 128 Hz ≈ 500 ms); ShallowConvNet uses 40 filters of 25 taps at 250 Hz (100 ms). Ours sits in between, with more filters.

3–4. Spatial convolution (64 filters spanning all 63 electrodes) + BatchNorm + ELU.

- Kernel `(63, 1)`: each output is a **weighted sum over all electrodes** at one time point. This is a learned **spatial filter**, the deep-learning version of classical techniques like Common Spatial Patterns (CSP), beamformers, or picking an ICA component: it can, for example, weight the occipital electrodes positively and subtract a frontal reference.
- `groups=1` means each of the 64 spatial filters mixes **all 32 temporal maps**: it combines frequency bands and electrodes jointly ($32 \times 63 = 2{,}016$ weights per filter). EEGNet instead uses a **depthwise** spatial conv (each spatial filter sees only one temporal map, `D = 2` spatial filters per temporal filter, 63 weights each, with a max-norm constraint). Ours has many more parameters here (129k vs EEGNet's 1,008 at the same input size). *Note:* the original scaffold's comment calls this layer "depthwise" but the code has always used `groups=1` (commit `db3fb74`); the code is what ran in every experiment.
- The kernel height is the channel count, so **the layer is tied to 63 electrodes in a fixed order**. That's why a THINGS-EEG2 model can't run on the final phase's 32 Emotiv channels without retraining (see issue #21 and the planned position-based spatial attention).
- **ELU** (exponential linear unit, Clevert et al., 2016) is smooth for negative inputs; EEGNet and ShallowConvNet-family models use it.

5. Average pooling (×4) + Dropout.

- 120 → 30 time steps: an average over 4 samples is a crude **low-pass filter followed by downsampling** to 30 Hz (33 ms per step). Evoked visual responses are slow (most power below ~20 Hz), so little is lost and later layers become cheaper and see longer stretches of time.
- **Dropout 0.25** randomly zeroes 25% of activations during training (Srivastava et al., 2014). EEGNet's paper recommends 0.25 for cross-subject training (0.5 within-subject) **(from memory of the EEGNet paper; braindecode's default is 0.25)**.

6–8. Depthwise-separable convolution + BatchNorm + ELU.

- **Depthwise** (layer 6): each of the 64 maps gets its **own** 15-tap temporal filter (`groups=64`, 960 weights). At 30 Hz, 15 taps span **500 ms**: it summarises how each spatial component evolves over half a second.
- **Pointwise** (layer 7): a 1×1 conv that mixes the 64 maps into 128 (8,192 weights).
- Splitting "filter in time" from "mix the maps" is a **depthwise-separable convolution** (Chollet, Xception, CVPR 2017), adopted by EEGNet to use far fewer parameters than a full conv (a full 64→128 conv with 15 taps would need 122,880 weights instead of 9,152).

9. Average pooling (×4) + Dropout. 30 → 7 time steps (30/4 = 7.5, floored: the last two 30 Hz steps are not pooled, though their information still reaches step 6 through the depthwise kernel). Each final step represents 16 input samples = **133 ms**.

10. Flatten → 896 features. 128 maps × 7 time steps. Flattening (rather than averaging over time) keeps **when** each feature occurred: the next layer has separate weights for "component 12 at 100–233 ms" and "component 12 at 500–633 ms". That matters because the timing of visual responses carries information. It is also our best explanation for why random time-shift augmentation hurt (R4, §5).

Receptive field of the final features. Following the layers back to the input: temporal conv (31) → pool (stride 4) → depthwise conv (15 at stride 4) → pool (stride 16). Each of the 7 final time steps sees **102 input samples = 850 ms**, and consecutive steps are **133 ms apart**. Step 0 covers roughly −200 → +283 ms around image onset; step 6 roughly +242 → +792 ms. So every final feature integrates most of the window, with a different centre of gravity: early steps emphasise the first visual responses (~100–200 ms), late steps the later, more semantic activity.

11–13. Projection head. `Linear(896→512) → BatchNorm1d → GELU → Dropout → Linear(512→1536).`

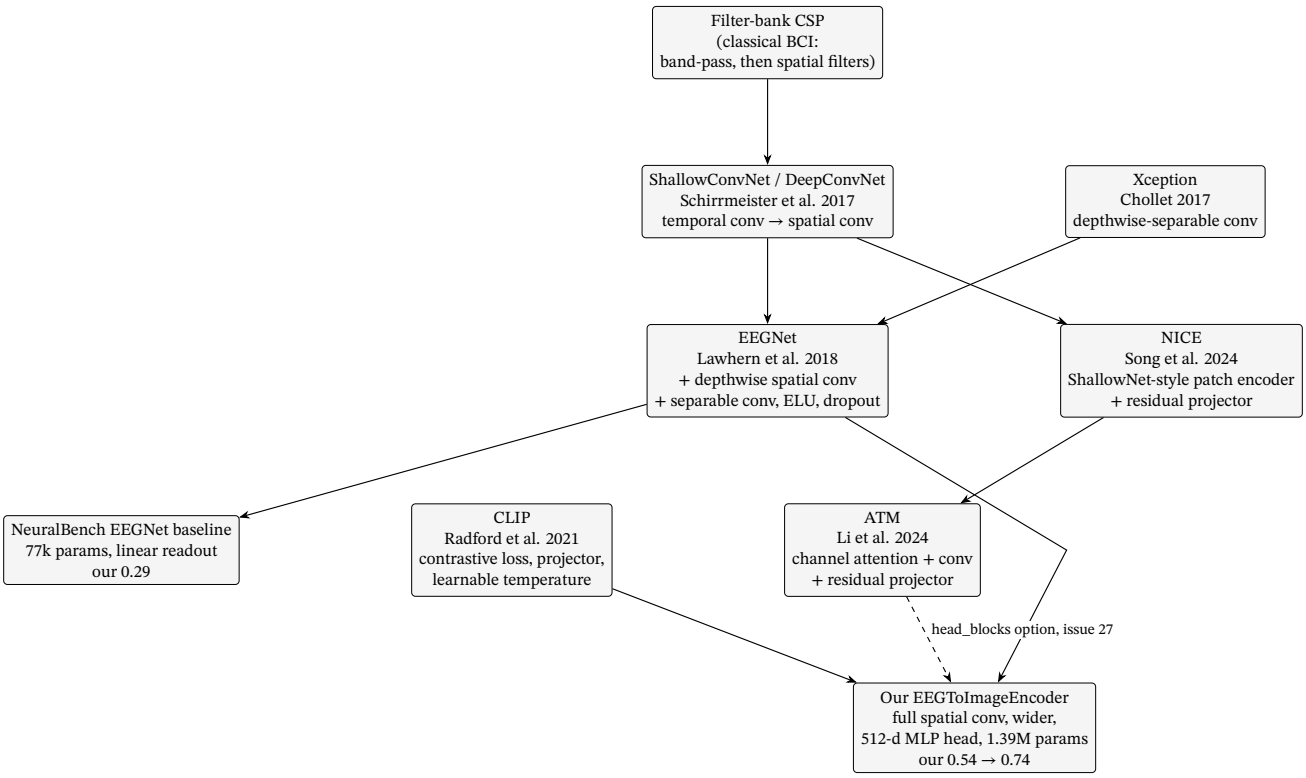
- A two-layer MLP that maps EEG features into DINOv2's 1536-d space. Using an MLP "projector" between the encoder and the contrastive loss is the standard recipe in contrastive learning (CLIP uses linear projections; SimCLR showed a non-linear head helps).
- **GELU** (Hendrycks & Gimpel, 2016) is a smooth ReLU used in transformers.
- It's 90% of the model. The NeuralBench EEGNet baseline instead reads 48 features (16 maps × 3 time steps) straight into 1536 outputs with a single linear layer; that "bigger head" is one of the two changes `RESULTS.md` credits for our jump from 0.30 to 0.54 (§2.2).
- Optional (#27): `head_blocks` adds residual blocks `x + Dropout(Linear(GELU(x)))` on the 1536-d output followed by `LayerNorm`, as in ATM's projector (Li et al., 2024). Off by default; R13/R14 in the queue test it.

14. L2 normalisation. Divides the output by its length so it lies on the unit sphere. The metric ranks by **cosine similarity**, which only depends on direction, and the loss normalises the targets too, so predictions and targets are compared purely by angle.

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2. Where the encoder comes from

2.1 Family tree



- **Filter-bank CSP** is the classical pipeline for EEG decoding: band-pass filter the signal into several frequency bands, learn spatial filters per band, then classify the log-power.
- **ShallowConvNet** (Schirrmeister et al., 2017) turned that into a network: temporal conv (40 filters × 25 taps) → spatial conv (40) → squaring → mean pooling (75, stride 15) → log → classifier, i.e. a learnable filter-bank CSP. **DeepConvNet** stacks four conv-pool blocks. (Defaults verified in braindecode's `ShallowFBCSPNet`: 40 / 25 / 40 / 75 / 15.)
- **EEGNet** (Lawhern et al., 2018) made it compact: temporal conv (F1 = 8 filters, 64 taps) → depthwise spatial conv (D = 2 per temporal filter, max-norm constrained) → BatchNorm, ELU, average pooling, dropout → separable conv (F2 = 16) → pooling → classifier. (Defaults verified in braindecode's `EEGNet`: F1 = 8, D = 2, F2 = F1·D, kernel 64, depthwise kernel 16, dropout 0.25.)
- **Our encoder** keeps EEGNet's block order and its separable conv, but uses 4× more temporal filters (32), a **full** (non-depthwise, unconstrained) spatial conv with 64 filters, 128 separable maps, a shorter temporal kernel (31 taps at 120 Hz), and replaces the classifier with a 2-layer **projection head** into DINOv2 space trained with a CLIP loss.
- **NICE** (Song et al., ICLR 2024) and **ATM** (Li et al., NeurIPS 2024) are the THINGS-EEG2 state of the art that issue #12 targets. From the papers and their public code, roughly **(unverified in detail; check the papers' architecture figures)**: NICE's encoder is a ShallowNet-style temporal conv → average pooling → spatial conv producing a short sequence of patch embeddings, then a projector `Linear → residual(GELU, Linear, Dropout) → LayerNorm`; NICE-GA adds graph attention over electrodes. ATM adds a channel-wise transformer embedding (with a subject token) before a similar conv block and residual projector. Our `head_blocks` option is that residual projector.

2.2 The true history (from git)

Date	Commit	What happened
2026-09-11	`db3fb74`	Initial scaffold. `src/models/eeg_encoder.py` defines the architecture as **learning TODOs** "based on EEGNet principles": temporal conv (1, 31) × 32 → spatial conv (n_chans, 1) × 64 → separable conv → 512-d projection head → 1536, L2-normalised. The layer sizes we still use were fixed here.
2026-09-13	`7d7dc1e`	TODOs completed ("complete TODOs for setup").
2026-10-17/18	`6896932`, `bf5bf17`	`submission/submission.py` made self-contained (torch only) and fixed for the Codabench contract; state_dict keys match `src/models/eeg_encoder.py`.
2026-10-18	`7b439e0`	**First real upload: NeuralBench's own EEGNet** (official template), 0.29.
2026-10-19	`da20e91`, `c652e3a`	Our encoder trained on THINGS-EEG2 with InfoNCE: **0.54** (held-out 42.15%). Seeded training and `weights_*.pt` ensembling added.
2026-10-19	`5facb9b`	Training options for the round-1 experiments (`learn_temperature`, `dedup`, `symmetric`, augmentation) and `--set` overrides.
2026-10-19	`1a8fc30`, `41b152e`, `2fb5a51`	Learnable temperature (+10 held-out), `--use-val` (+3.5).

Date	Commit	What happened
2026-10-20	`e34d3f1`	Final 3-seed ensemble on all concepts: 73.80% test, **0.74** on Codabench.
2026-10-20	`18992e0`, `424c9b5`	Per-epoch held-out scoring + early stopping (#15); repetition averaging (#17).
2026-10-20	`e893e30`, `c5881bd`, `48925dd`	Schedule / weight decay / dropout settings (#18); encoder size settings, sizes read back from weights (#27); `min_temperature`.
2026-10-20	`68d2b65`, `0e9ff1f`	Window cache (#14); per-epoch checkpoints and `--resume`; pause/resume.

Why our encoder beat the NeuralBench baseline by 24 points (`RESULTS.md`: "bigger head, 10× higher lr"). The baseline's configuration (`submissions/neuralbench_eegnet_things_eeg2_seed34/run_config.yaml`) versus ours:

	NeuralBench EEGNet (0.29–0.30)	Our encoder (0.54)
Network	braindecode EEGNet defaults (F1 8, D 2, F2 16, kernel 64), **77k parameters** (counted from the shipped weights)	1.39M parameters
Readout	one linear layer: 48 features (16 maps × 3 steps) → 1536	896 features → 512 → 1536 MLP
Optimiser	AdamW, **lr 1e-4**, weight decay 0.05, OneCycle schedule	AdamW, **lr 1e-3**, weight decay 1e-4, constant
Batch	64	256 (4× more negatives per step)
Loss	NeuralBench `ClipLoss`, fixed temperature	our `clip_loss`, $\tau = 0.07$ fixed (learnable later)
Stopping	40 epochs, patience 5 on a validation rank metric	20 epochs

We never ran the ablation that separates these factors, so the split between "bigger head", "higher lr" and "bigger batch" is **not measured**.

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3. The loss and the training recipe

All in `Solver.fit` and `clip_loss` in `submission/submission.py`, driven by `scripts/train_solver.py`.

3.1 The contrastive (InfoNCE / CLIP) loss

For a batch of B windows, let p_i be the model's unit-length prediction for window i and t_i the DINOv2 embedding of the image that window shows (normalised inside the loss). The similarity matrix is $s_{ij} = p_i \cdot t_j$ (cosine). With temperature τ:

$$\mathcal{L} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp(s_{ii}/\tau)}{\sum_{j=1}^{B} \exp(s_{ij}/\tau)}$$

Read it as a **B-way classification per row**: "which of the B images in this batch did EEG window i see?" The correct answer is the diagonal; the other $B-1$ images in the batch are the **negatives** ("in-batch negatives"), so no extra sampling is needed. Minimising it pulls p_i towards t_i and pushes it away from the other targets, which is exactly what top-k retrieval rewards. This is **InfoNCE** (van den Oord et al., 2018), popularised by **CLIP** (Radford et al., 2021), and used for brain decoding by Défossez et al. (2023), NICE, ATM.

Why not regression (minimise $\|p_i - t_i\|^2$)? A regression model facing noisy input predicts the conditional mean, a blurry "average image" that sits close to everything and ranks poorly. #23's post-hoc ridge maps confirmed it: fitted by squared error, they scored 0.046–0.052 held-out (§5).

Options tested in round 1:

- `symmetric=True` adds the column-wise term (for each image, which EEG window matches?), as in CLIP. No gain here (R3).
- `dedup=True` masks out "false negatives": other windows in the batch that show the **same** image (each training image has ~40 windows: 10 subjects × 4 repetitions). With ~14.5k training images and batch 256, a batch contains only about 2 such pairs on average **(our estimate)**, which explains why it made no difference (R1: 41.20% vs 41.35%).

3.2 The temperature

τ divides the similarities before the softmax. Small τ makes the softmax sharp, so the loss concentrates on the hardest negatives (images that look most like the right one).

- Fixed at **0.07** at first (a common default).
- **Learnable** (`learn_temperature=True`), parameterised as CLIP does: the trainable parameter is $\log(1/\tau)$ (initialised to $\log(1/0.07) \approx 2.66$) and it is clamped at $\log 100$, i.e. $\tau \geq 0.01$ ("logit scale at most 100"). It's trained with the model by the same optimiser.
- Effect: **+10 points held-out** (R2: 41.35% → 51.75%), the biggest single gain.
- In R6 the learned τ fell from 0.027 (epoch 1) to the **0.01 floor by epoch ~15** and stayed there. R15 lowered the floor to 0.005 (`min_temperature`): τ went to 0.0072, but held-out trailed R6 at every comparable epoch (best 0.564 vs 0.582 by epoch 31) when a restart stopped it at epoch 32. **Conclusion: no gain**, floor stays at 0.01.

3.3 Optimisation

Setting	Value	Notes / source
Optimiser	**AdamW**	Adam with decoupled weight decay (Loshchilov & Hutter, ICLR 2019)
Learning rate	**1e-3**	10× the NeuralBench baseline
Weight decay	**1e-4**	`weight_decay` setting (#18); applied to the temperature parameter too (negligible)
Batch size	**256**	`--batch-size`; more negatives per step. R10/R11 test 512 / 1024 with linearly scaled lr (Goyal et al., 2017)

Setting	Value	Notes / source
Gradient clipping	max norm **1.0**	stabilises early training
Schedule	constant (default); `lr_schedule='cosine'` + `warmup_epochs`	Cosine decay from SGDR (Loshchilov & Hutter, ICLR 2017); linear warm-up from Goyal et al. 2017. Applied per step via `LambdaLR`
Epochs	20 originally; **40** in R6 (+3.75 held-out)	longer training still improving at 40
Dropout	0.25	`dropout` setting
Seed	`--seed`	seeds weight init, shuffling and dropout

3.4 Data-side recipe

- **More data (`--use-val`, #16):** NeuralBench's own "val" split of THINGS-EEG2 is ordinary training data (same concepts). Adding it gives +25% windows (470,140 → 587,574 with the held-out concepts removed; 668,400 for final models on all concepts): **+3.5 held-out**.
- **Augmentation (R4):** random circular time shift up to ±6 samples (±50 ms) and 10% channel dropout. It **hurt badly** (49.85% → 36.80%). Our interpretation **(hypothesis, not tested)**: the flattened readout uses absolute timing (§1.3, layer 10), the circular roll wraps the end of the window onto the start, and 50 ms is large relative to visual response latencies.
- **Repetition averaging (`--avg-reps K`, #17, Gifford et al. averaged repetitions too):** each training item becomes the mean of K repetitions of the same (subject, image), which cuts independent noise by about $\sqrt{K}$. With `--avg-reps-random` K is drawn from 1..K per item, mixing clean and noisy examples (test inputs are single trials). There are 145,400 (subject, image) groups, so an "epoch" is **4× shorter** than with single windows. **Caveat:** early stopping counts epochs, so patience 5 is effectively 4× less patient. R7 (random 1–4) was still rising when it stopped at 0.555 (R6: 0.590), but it got there in 31 min, where R6 needed ~105 min for 0.59 and was at ~0.51 after 31 min. A fair test needs patience scaled to the epoch length.

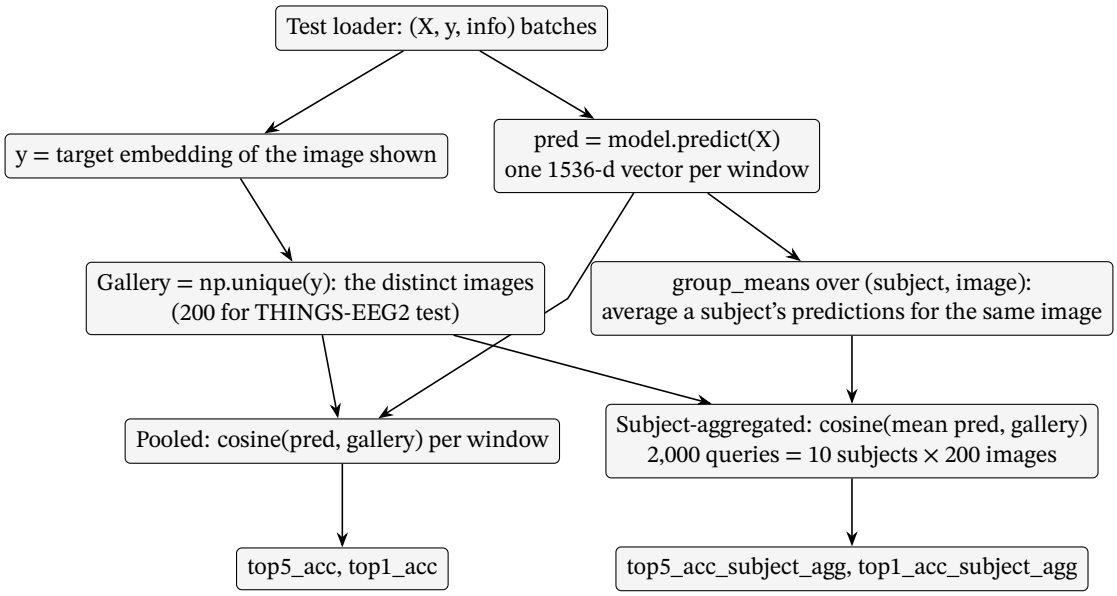
3.5 Training infrastructure (no effect on results, by design)

- **Early stopping on held-out (#15):** after each epoch, `scripts/train_solver.py` scores the held-out split with the official objective; `fit` keeps the best epoch's weights and stops after `patience` epochs without improvement (5; 8 for cosine runs). Caveat for teaching: picking the best epoch on held-out makes the reported held-out score slightly optimistic (a mild selection bias); comparisons between runs remain fair because every run gets it.
- **Window cache (#14, `scripts/cache_windows.py`, `--cache`):** windows extracted once to memory-mapped `.npy` files (24 GB). Verified to give bit-identical windows, scores and training batches, with ~4× faster loading and much less memory.
- **Checkpoints and `--resume`:** `fit` saves its full state (model, optimiser, scheduler, temperature, best epoch, history, RNG states) after every epoch; a resumed run on CPU ends with bit-identical weights. `make pause-training` / `resume-training` freeze and continue a run (used when the laptop's GPU is needed for the screen).

4. Evaluation in detail

4.1 The official objective

`external/2026-competition/tracks/image_decoding/objective.py`, `Objective.evaluate_result`:



- **Gallery:** the unique target embeddings of the split, found with `np.unique(y_true, axis=0)`. Repeated presentations of an image share an identical embedding, so they collapse to one candidate.
- **Scores:** predictions and candidates are L2-normalised; the score is the dot product (cosine similarity).
- **Top-k accuracy** (`benchmark_utils/metrics.py`, `topk_accuracy`): a query is a hit if its true image's rank is below k. Ties share their average rank. **Chance level**: 5/200 = **2.5%** for top-5, 0.5% for top-1.
- **Subject-aggregated** (`group_means`): average the predictions of all windows with the same (subject, image) key, then rank those means. On the test set each subject saw each image ~80 times, so this averages ~80 predictions per query: it's why our top-5 is 0.74 subject-aggregated but 0.31 pooled. The sealed phase ranks on top-5 subject-aggregated; the warm-up leaderboard's main column has matched our subject-aggregated top-5 on every upload. (The objective's docstring says warm-up uses the pooled metric; the leaderboard columns show both, and README/RESULTS track subject-aggregated. **Worth confirming with the organisers.**)

4.2 Our held-out-concept split

`src/data/heldout.py` + `configs/heldout_concepts.json` (seed 0, committed):

- 200 of the 1,654 training **concepts** are removed from training entirely (no window of those concepts is trained on), so the model must generalise to unseen concepts, like the test set.
- One image per held-out concept (e.g. `air_conditioner_06s`) forms a **200-image gallery**, the same size as the test gallery; its windows (from NeuralBench's train and val splits) are the queries: 8,076 windows ≈ 200 images × 10 subjects × ~4 repetitions.
- Training then uses the remaining 1,454 concepts (587,574 windows with `--use-val`).

Why the held-out score is lower than the test score for the same model: the held-out images were shown only **~4 times** per subject (training-set protocol) while test images were shown **~80 times**. Subject aggregation averages ~4 predictions instead of ~80, so much less noise is removed. The ranking of models is what matters, and it has been consistent.

4.3 From held-out to the leaderboard (empirical)

Run	Held-out top-5 SA	Test top-5 SA (same model)	Gap
control (seed 1)	0.4135	0.5450	+13.2
R2 learnable τ	0.5175	0.6525	+13.5
R5 + val data	0.5525	0.7110	+15.9

Then retraining the chosen recipe on **all** concepts with 3 seeds and ensembling added +2.7 on test (R5 0.7110 → 0.7380). So a held-out gain of x points has translated into roughly x points on the leaderboard. R6's 0.590 suggests about **0.75–0.78** for its final ensemble **(a forecast, not a measurement)**.

Why local test = Codabench: same data pipeline (NeuralBench pinned to the worker's version, `neuralbench==0.4.0` etc.), same test split, same objective code, deterministic inference. All six uploads so far matched local scores to two decimals.

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5. Every experiment and its result

Held-out = top-5 subject-aggregated on our held-out split (seed 1 unless noted). Test = top-5 subject-aggregated on THINGS-EEG2 test, computed locally. Codabench = leaderboard score. Sources: `RESULTS.md`, `logs/overnight_summary.md`, `logs/round2.status`, `logs/*.log`, `submissions/*/run_config.json`.

Run	Hypothesis	Change	Held-out	Test	Cod-abench	Conclusion
Neural-Bench EEGNet, seed 34	Official baseline as a floor	braindecode EEGNet, NeuralBench recipe	–	0.288	**0.29**	Starting point (submission 1001528)
EEGNet ensemble 34+35	Seed averaging helps	mean of 2 models' normalised predictions	–	0.2975	0.30	+1; 3 seeds tied (0.2970)
REVE frozen + ridge	A foundation model's features suffice	official ridge probe on frozen REVE	–	0.025	–	Chance level
REVE frozen + linear CLIP head	Same, with a contrastive head	linear head, CLIP loss	–	0.127 (120 Hz) / 0.086 (200 Hz)	–	Far below EEGNet; deprioritised
Our encoder (unseeded)	Bigger model + higher lr + CLIP loss	§2.2 table, τ 0.07, 20 epochs	0.4215	0.5415	**0.54**	+24 over baseline
3-encoder ensemble	Seeds help	3 seeds	0.4315	0.5530	0.55	+1

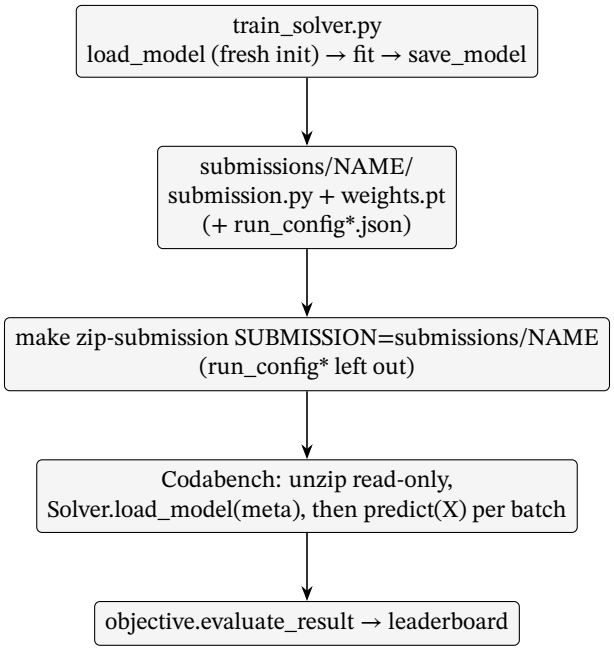
Run	Hypothesis	Change	Held-out	Test	Cod-abench	Conclusion
Control (seed 1)	Reference for round 1	same, seed 1	0.4135	0.5450	–	
R1 dedup	False negatives hurt	`dedup=True`	0.4120	–	–	No gain (few duplicates per batch)
R2 learnable τ	$\tau = 0.07$ is too soft	`learn_temperature=True`	**0.5175**	0.6525	**0.65**	**+10.4**: keep
R3 τ + dedup + symmetric	CLIP's symmetric loss helps	+ `dedup`, `symmetric`	0.4985	–	–	Worse than R2: drop
R4 R3 + augmentation	Invariance to shifts/channels	$\pm$50 ms circular shift, 10% channel dropout	0.3680	–	–	Hurts a lot: drop
R5 R2 + val data	More data helps	`--use-val` (+25%)	**0.5525**	0.7110	**0.71**	**+3.5**: keep
#23 recalibration	Post-hoc linear map fixes predictions	centering / ridge maps on R2's outputs	0.376 / 0.046–0.052 (none: 0.5175)	–	–	All hurt: closed
Final 3-seed ensemble	All concepts + ensemble	R5 recipe on all 668,400 windows, seeds 1–3	n/a	**0.7380** (single seeds 0.715–0.721)	**0.74** (top-1 0.40)	Current best (submission 1004941)
R6 40 epochs	20 epochs were too few	`n_epochs=40`, early stopping	**0.590** (epoch 36)	not run	–	**+3.75**: keep; still rising at 40
R15 floor 0.005	τ is stuck at the 0.01 floor	`min_temperature=0.005`	0.564 at epoch 31 (stopped at 32/40 by a restart)	–	–	Trailed R6 throughout: no gain
R7 avg 1–4 reps (random)	Cleaner training examples	`--avg-reps 4 --avg-reps-random`	0.555 (epoch 46, stopped at 51)	–	–	Cut short by patience (§3.4); faster per minute than R6; re-test with scaled patience
R8 avg 2 reps	Same, fixed k = 2	`--avg-reps 2`	running at the time of writing (0.508 at epoch 28)	–	–	–

Top-1 subject-aggregated for the uploaded models: 0.09 (EEGNet) → 0.22 → 0.34 (R2) → 0.37 (R5) → **0.40** (final ensemble).

Queued (`logs/round2.status`): R9–R11 cosine schedule with batch 256 / 512 / 1024 (#18); R12–R14 wider convolutions / head 1024 + residual block / both (#27).

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6. Submission and inference path



- **Contract** (`benchmark_utils.base_solver.CompetSolver`): Codabench never trains. It calls `Solver.load_model(meta)` with `n_chans`, `n_times`, `n_outputs`, `ch_names`, `chs_info`, `device`, `submission_dir`, and then `model.predict(X)` on `(B, C, T)` batches; `predict` must return `(B, n_outputs)`.
- **Inference vs training in `load_model`:** if `train_loader` is set (a local training run) it builds a fresh, seeded model; otherwise it loads the shipped weights.
- **Sizes from the weights (`encoder_sizes`, #27):** the number of filters, the head width and the number of residual blocks are read from the tensor shapes in the weights file, so the uploaded `submission.py` never needs editing after a wider run.
- **Ensembles (`EmbeddingEnsemble`):** if the folder contains `weights_*.pt` files, one model is built per file and their (unit-length) predictions are averaged.
- **Run time on Codabench:** 226–318 s per upload so far (limit 60 min), e.g. 305 s for the 3-model ensemble.
- **Reproducibility:** each run writes `run_config.json` (command, commit, settings, seed, windows, minutes, best epoch) and `run_config_curve.json` (per-epoch loss, temperature, lr, held-out), which the top-3 audit needs.

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7. Paper map and bibliography

7.1 Idea → paper → where in our code

Idea in our code	Paper	What to read	Where it lives
Temporal-then-spatial conv factorisation	Schirrmeister et al. 2017	ShallowConvNet / DeepConvNet architecture figures and their FBCSP motivation	`EEGToImageEncoder.temporal_conv`, `.spatial_conv`
Compact EEG CNN: block order, ELU, pooling, dropout, separable conv	Lawhern et al. 2018 (EEGNet)	Architecture section and table of layers	`EEGToImageEncoder` (whole conv part)
Depthwise-separable convolution	Chollet 2017 (Xception)	Section on depthwise-separable convs	`EEGToImageEncoder.sep_conv`
BatchNorm	Ioffe & Szegedy 2015		every `BatchNorm*`
ELU / GELU activations	Clevert et al. 2016 / Hendrycks & Gimpel 2016		`spatial_conv`, `sep_conv` / `projection_head`
Dropout	Srivastava et al. 2014		`Dropout(dropout)`
Contrastive loss with in-batch negatives	van den Oord et al. 2018 (InfoNCE); Radford et al. 2021 (CLIP)	CLIP §2.3 and its Figure 3 pseudo-code	`clip_loss`
Learnable temperature, logit scale ≤ 100	Radford et al. 2021 (CLIP)	CLIP §2.5 (training details)	`Solver.fit` (`log_scale`, `min_temperature`)
Symmetric loss	Radford et al. 2021	Figure 3	`clip_loss(symmetric=True)`
CLIP loss for brain signals, retrieval evaluation	Défossez et al. 2023; Benchetrit et al. 2024	Methods (loss, retrieval metric)	`clip_loss`, objective
Zero-shot EEG→image retrieval on THINGS-EEG2	Song et al. 2024 (NICE); Li et al. 2024 (ATM)	Problem setup and results tables	whole project
Residual projector + LayerNorm	Li et al. 2024 (ATM); NICE's projector	Architecture section	`ResidualBlock`, `head_blocks`
DINOv2-giant target embeddings	Oquab et al. 2023 (DINOv2)	Intro + feature quality sections	targets (NeuralBench, `facebook/dinov2-giant`)
AdamW	Loshchilov & Hutter 2019		`Solver.fit` optimiser
Cosine schedule	Loshchilov & Hutter 2017 (SGDR)		`Solver._lr_factor`
Warm-up, lr scaling with batch size	Goyal et al. 2017	§2.1–2.2	`_lr_factor`, R10/R11
Repetition averaging / data protocol	Gifford et al. 2022 (THINGS-EEG2)	Paradigm and preprocessing	`src/data/repetitions.py`

Idea in our code	Paper	What to read	Where it lives
Concepts and images	Hebart et al. 2019 (THINGS)		`configs/heldout_concepts.json`
Blurred-image targets (issue #26)	UBP (CVPR 2025)		not implemented yet
Position-based spatial attention, spatial dropout	Défossez et al. 2023	"Spatial attention" layer in Methods (Fourier embedding of sensor positions)	`SpatialAttention`, `Solver.virtual_channels` (§9.2)
Electrode positions (10-05 system)	Oostenveld & Praamstra 2001 **(unverified)**; MNE's `standard_1005` montage		`ELECTRODE_XY`, `channel_positions`
Held-out participants (cross-subject validation)	standard practice; motivated by the sealed final's unseen subjects		`heldout_subjects_split` (§9.1)
EEG foundation model (tried, weak)	El Ouahidi et al. 2025 (REVE)		`logs/reve*`
Final-phase data family	Xu et al. 2024 (Alljoined1); Xu et al. 2025 (Alljoined-1.6M)	Dataset papers	issues #4, #21
A convolution + transformer EEG encoder (candidate for #12)	Song et al. 2023 (EEG Conformer)	Architecture figure	not implemented

7.2 Bibliography

Verification: "verified" means the citation was checked against metadata shipped in our installed packages (braindecode, NeuralFetch, PyTorch, transformers). Web search was not available while writing this, so the others are from memory and marked **(unverified)**.

Datasets

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- Hebart, M. N., et al. (2019). THINGS: A database of 1,854 object concepts and more than 26,000 naturalistic object images. *PLoS ONE* 14(10): e0223792. **(unverified)**
- Xu, J., Aristimunha, B., Feucht, M. E., Qian, E., et al. (2024). Alljoined — A dataset for EEG-to-Image decoding. arXiv:2404.05553 — verified (NeuralFetch `xu2024alljoined`; 8 subjects).
- Xu, J., Nunes, U. B., Jiang, W., Ryther, S., Pringle, J., Scotti, P. S., Delorme, A., & Kneeland, R. (2025). Alljoined-1.6M: A million-trial EEG-image dataset for evaluating affordable brain-computer interfaces. arXiv:2508.18571 — verified (NeuralFetch `xu2025alljoined`; 20 subjects, Emotiv headset).

EEG architectures

- Lawhern, V. J., Solon, A. J., Waytowich, N. R., Gordon, S. M., Hung, C. P., & Lance, B. J. (2018). EEGNet: a compact convolutional neural network for EEG-based brain–computer interfaces. *Journal of Neural Engineering* 15(5), 056013. arXiv:1611.08024 — verified (braindecode).

- Schirrmeister, R. T., Springenberg, J. T., Fiederer, L. D. J., Glasstetter, M., Eggensperger, K., Tangermann, M., Hutter, F., & Ball, T. (2017). Deep learning with convolutional neural networks for EEG decoding and visualization. *Human Brain Mapping*. doi:10.1002/hbm.23730 — verified (braindecode).
- Song, Y., Zheng, Q., Liu, B., & Gao, X. (2023). EEG Conformer: Convolutional transformer for EEG decoding and visualization. *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 31, 710–719 — verified (braindecode; braindecode dates it 2022, the journal volume is 2023).
- Chollet, F. (2017). Xception: Deep learning with depthwise separable convolutions. *CVPR* — verified (cited in braindecode's EEGNet).
- El Ouahidi, Y., Lys, J., Thölke, P., Farrugia, N., Pasdeloup, B., Gripon, V., Jerbi, K., & Lioi, G. (2025). REVE: A foundation model for EEG — adapting to any setup with large-scale pretraining on 25,000 subjects. *NeurIPS 2025*. openreview.net/forum?id=ZeFMtRBy4Z — verified (braindecode).

EEG/MEG → image decoding

- Song, Y., Liu, B., Li, X., Shi, N., Wang, Y., & Gao, X. (2024). Decoding natural images from EEG for object recognition (NICE). *ICLR 2024*. arXiv:2308.13234 **(unverified)**
- Li, D., Wei, C., Li, S., Zou, J., & Liu, Q. (2024). Visual decoding and reconstruction via EEG embeddings with guided diffusion (ATM). *NeurIPS 2024*. arXiv:2403.07721 **(unverified)**
- "Bridging the vision-brain gap with an uncertainty-aware blur prior" (UBP). *CVPR 2025*. **(unverified: authors and arXiv ID; search the title)**
- Benchetrit, Y., Banville, H., & King, J.-R. (2024). Brain decoding: toward real-time reconstruction of visual perception. *ICLR 2024*. arXiv:2310.19812 **(unverified)**
- Défossez, A., Caucheteux, C., Rapin, J., Kabeli, O., & King, J.-R. (2023). Decoding speech perception from non-invasive brain recordings. *Nature Machine Intelligence* 5(10), 1097–1107 — verified (braindecode REVE references). arXiv:2208.12266 **(ID unverified)**

Representation learning and optimisation

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- van den Oord, A., Li, Y., & Vinyals, O. (2018). Representation learning with contrastive predictive coding. arXiv:1807.03748 **(unverified)**
- Oquab, M., et al. (2023). DINOv2: Learning robust visual features without supervision. arXiv:2304.07193 — verified (transformers' DINOv2 module).
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8. Teaching kit

8.1 A 60–90 minute session

Time	Topic	Show
0–10 min	The task and why it's retrieval, not classification; chance level 2.5%	how_it_works.md, first diagram
10–20	The data: THINGS-EEG2, windows, why repetitions matter (4 vs 80)	how_it_works.md §2; §4.2 here
20–35	The encoder, layer by layer: temporal → spatial → separable → head; receptive field	§1.1 diagram and §1.2 table
35–50	The loss: the B×B matrix, negatives, temperature; why not regression	§3.1 formula; how_it_works.md loss diagram
50–60	Evaluation and the held-out split; why held-out < test	§4.1 diagram, §4.3 table
60–75	The experiment log: what worked (τ, data, epochs), what didn't, and why	§5 table
75–90	What's next: final phase = 32 channels + unseen people; headset-independent spatial layer; held-out participants	how_it_works.md §8

Tip: before the session, run exercise 2 so the scores appear live on screen.

8.2 Check-your-understanding questions (with answers)

1. **Why can the model recognise images of concepts it never trained on?** It predicts a point in DINOv2's embedding space, not a class. DINOv2 places any image, including unseen ones, in that space, so retrieval works for new images (zero-shot).
2. **What is chance-level top-5 accuracy on the test set?** $5/200 = 2.5\%$.
3. **Why is subject-aggregated accuracy (0.74) so much higher than pooled (0.31)?** It averages ~80 predictions per (subject, image) before ranking, which cancels most of the single-trial noise.
4. **What does the temporal convolution learn, in signal-processing terms?** 32 FIR filters of 258 ms shared across electrodes: band-pass filters or response-shape detectors.
5. **Why does the model have to be retrained for the 32-channel final phase?** The spatial conv's kernel spans exactly 63 electrodes in a fixed order; its weights have no meaning for a different montage.
6. **Where are 90% of the parameters?** In the projection head (896 → 512 → 1536).
7. **Write the InfoNCE loss for one row and say what the negatives are.** $-\log \frac{e^{s_{ii}/\tau}}{\sum_j e^{s_{ij}/\tau}}$; the negatives are the other images in the same batch.
8. **What does lowering the temperature do?** Sharpens the softmax so the loss focuses on the hardest wrong images. Learning it gave +10 held-out; going below 0.01 gave nothing.
9. **Why did `dedup` make no difference?** Same-image duplicates are rare in a batch of 256 (about 2 pairs), so there were few false negatives to remove.
10. **Give one plausible reason augmentation by time shift hurt.** The flattened readout relies on absolute timing of responses; shifting by up to ±50 ms (with wrap-around) corrupts that.
11. **Why is the held-out score lower than the test score for the same model?** Held-out images were seen ~4 times per subject, test images ~80, so subject aggregation removes less noise on held-out.
12. **Why don't we choose models by their test score?** The test set is the public leaderboard data; selecting on it overfits to it and tells us nothing about the sealed phase. We select on the held-out split.
13. **Why do we retrain on all concepts for uploads?** More concepts cover DINOv2's space better; it added ~+2.7 on test together with the 3-seed ensemble.
14. **Why was R7 (repetition averaging) not a fair loss?** Its epochs are 4× shorter, so patience 5 stopped it while it was still improving.

15. **What does `encoder_sizes` solve?** It reads the layer sizes from the weights, so the same `submission.py` can load a model of any width without edits.

8.3 Hands-on exercises

Exercise 1: count the parameters and check the shapes (CPU, 1 minute).

```
# save as count_params.py at the repo root, run: .venv/bin/python count_params.py
import importlib.util, sys, torch
sys.path.insert(0, "external/2026-competition/tracks/image_decoding")
spec = importlib.util.spec_from_file_location("s", "submission/submission.py")
m = importlib.util.module_from_spec(spec); spec.loader.exec_module(m)
enc = m.EEGToImageEncoder(n_chans=63, n_times=120, output_dim=1536).eval()
x = torch.zeros(2, 1, 63, 120)
for block in (enc.temporal_conv, enc.spatial_conv, enc.sep_conv):
    for layer in block:
        y = layer(x)
        print(f"{str(layer)[:60]:60s} {tuple(x.shape)} -> {tuple(y.shape)} "
              f"{sum(p.numel() for p in layer.parameters())}")
        x = y
print("total:", sum(p.numel() for p in enc.parameters()))  # 1,387,872
```

Questions: where do 129,024 parameters come from? ($32 \times 63 \times 64$.) What changes with `EEGToImageEncoder(32, 120, 1536)`, the final phase's channel count?

Exercise 2: score an existing model like Codabench does (≈1 min with the cache).

```
python scripts/eval_local.py submissions/e6_r5_learntemp_useval --cache
 ↪  cache/windows/things_eeg2
# expect 0.7110 / 0.3745 / 0.2824 / 0.0958 (R5, Codabench 0.71)
python scripts/eval_local.py submissions/e6_r5_learntemp_useval --split heldout
 ↪  --cache cache/windows/things_eeg2
# expect 0.5525 held-out top-5 subject-aggregated
```

Questions: why is the second number lower? Which column is the leaderboard metric?

Exercise 3: plot learning curves (CPU, 2 minutes).

```
# save as plot_curves.py, run: .venv/bin/python plot_curves.py
import json, matplotlib.pyplot as plt
fig, (a, b) = plt.subplots(1, 2, figsize=(10, 4))
for run in ["m2_r6_epochs40", "m2_r7_avgreps4rand"]:
    curve = json.load(open(f"submissions/{run}/run_config_curve.json"))
    ep = [r["epoch"] for r in curve]
    a.plot(ep, [r["heldout"] for r in curve], label=run)
    b.plot(ep, [r["temperature"] for r in curve], label=run)
a.set(xlabel="epoch", ylabel="held-out top-5 (subject-agg)"); b.set(xlabel="epoch",
 ↪  ylabel="temperature", yscale="log")
a.legend(); plt.tight_layout(); plt.savefig("curves.png"); print("wrote curves.png")
```

Questions: when does the temperature reach its 0.01 floor? Re-plot with the x-axis in training samples (epochs × 587,574 for R6, × 145,400 for R7): which run learns faster per sample?

9. Next: new people and new headsets (in progress)

Everything above was tuned on THINGS-EEG2, where the test participants are the **same 10 people** the model trained on, recorded with **63 channels**. The sealed final is different: **11 new people, a 32-channel Emotiv headset, new images.** Two pieces of code (commits `25b12fb` and `d3ed4ea`, Oct 20) prepare for that. Neither has been trained at scale yet.

9.1 Measuring what the final measures: held-out participants

Our held-out split (§4.2) holds out *concepts* but not *people*, so it can't tell us how well a model works on someone it has never seen, which is notoriously the hard part of EEG (skull thickness, cortical folding and cap placement differ between people).

`train_solver.py --holdout-subjects` / `eval_local.py --split heldout_subjects` (`heldout_subjects_split` in `src/data/heldout.py`):

- the 200 held-out concepts are removed from training, as before;
- **participants 6 and 7** (`configs/heldout_subjects.json`, random with seed 0) are removed entirely: training uses 8 people;
- the score is computed on those 2 people's windows of the held-out concepts. Each held-out concept has 10 images, so there are **10 galleries of 200 images** (gallery g = the g-th image of every concept), each scored with the official metric, then averaged.

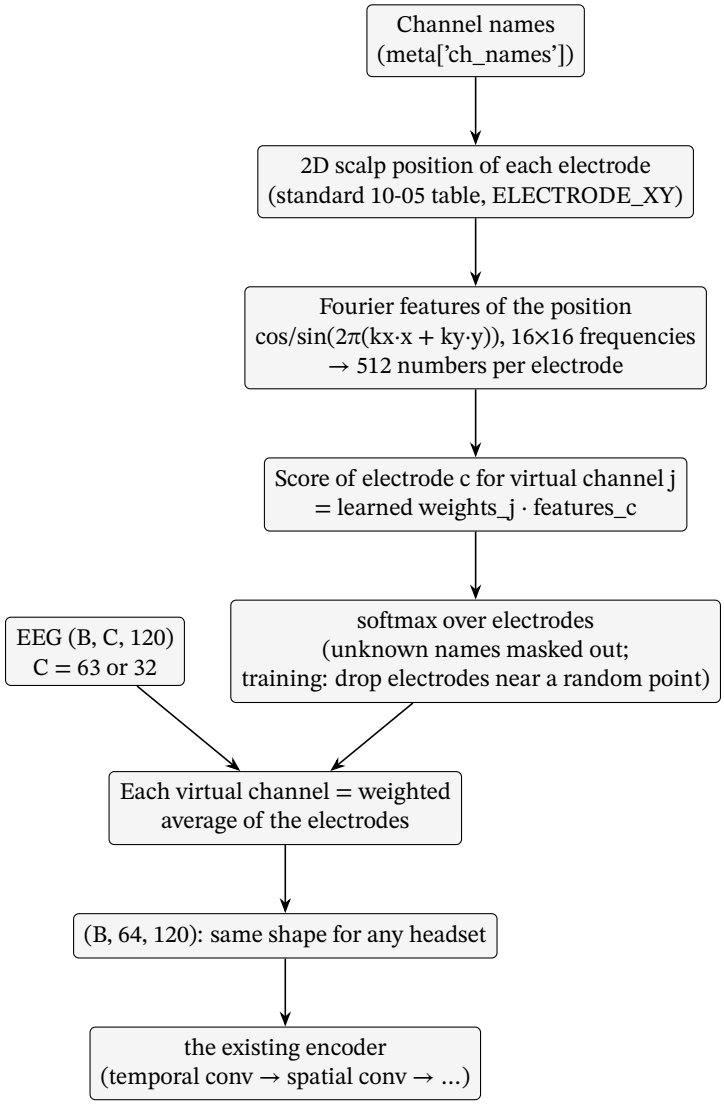
Why 10 galleries: one gallery with 2 people gives only 400 subject-aggregated queries of ~4 repetitions each (standard error ≈ 2.5 points, too noisy to judge 1–2-point changes). Ten galleries give 16,174 windows, ~4,000 queries, and a measured standard error of ≈ **0.8 points** (`eval_local` prints it as `top5_acc_subject_agg_se`).

Model	Trained on participants 6 and 7?	Held-out-participants score (top-5 SA)
R6	yes (reference, not a real held-out score)	0.6352 ± 0.0077
any `--holdout-subjects` run	no	to be measured: the drop from 0.635 is the cost of unseen people

9.2 One model for any headset: spatial attention

Our `spatial_conv` (§1.3) is a convolution with kernel height = number of channels, in a fixed order. Its weights only make sense for THINGS-EEG2's 63 channels, so a THINGS-EEG2 model can't read Emotiv's 32 channels, and the two datasets can't be trained together without remapping.

Défossez et al. (2023) solved this for MEG with a **spatial attention** layer, which we adapt (`SpatialAttention`, enabled with `--set virtual_channels=64`):



- **Why it's montage-independent:** the learned parameters (`heads`, $64 \times 512 \approx 33$k numbers) define a smooth *function of scalp position*, not one weight per channel. A new headset just evaluates that function at its own electrode positions.
- **Positions:** the pipeline only passes electrode *names*, and Codabench's worker only has `torch`, so `submission.py` embeds a table of 343 standard 10-05 positions, generated once from MNE's `standard_1005` montage (sphere-fitted, then flattened by an azimuthal-equidistant projection from the top of the head, like a topographic map), at a fixed scale, so 63- and 32-channel headsets share one coordinate frame.
- **Spatial dropout** (`spatial_dropout=0.15`): during training, every electrode within that radius of a random scalp point is ignored, so the model learns not to depend on any one region, which is useful when the final's headset lacks some of THINGS-EEG2's electrodes.
- **Checked** (synthetic data): with `virtual_channels=0` (the default) training is bit-identical to before; a model trained on 63 channels reloads to identical predictions and runs on a 32-channel Emotiv montage with the same weights. Positions are per-montage buffers, not saved in `weights.pt`.

9.3 The plan

1. Train the R6 recipe with `--holdout-subjects` (the unseen-people baseline).

2. Same with `--set virtual_channels=64 spatial_dropout=0.15`: does spatial attention cost anything on THINGS-EEG2, and does it help on unseen people?
3. Then joint training on THINGS-EEG2 + Alljoined (issue #21), possible only with step 2's layer, validated on Alljoined's held-out participants (issue #4).